WORKED SOLUTION

STEP I 1996, Question 13

Probability and Statistics • Discrete Distributions and Expectation

STEP I 1996, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q13

This question has a full guided treatment in *Beyond STEP*, Section 12.11, *Stamps, Cups and e*: https://beyondstep.keithmcnulty.org/ch12/stamps-and-e.html

The key idea

The game stops at the first wrong guess, so $N = k$ means "right with $2, 3, \ldots, k-1$ cups, then wrong with k cups". Multiplying these probabilities gives a factorial in the denominator, and factorials in a sum are a strong hint that the exponential series $e^x = \sum x^n/n!$ is about to appear. The trick for the expectations is to cancel factors of k against $k!$ so that each sum becomes a shifted copy of $\sum 1/n!$.

Solution

The distribution of N

With c cups jumbled, Jim is effectively guessing at random, so he is right with probability $1/c$ and wrong with probability $1 - 1/c$. Each round uses a fresh jumble, so the rounds are independent.

Since there are always at least two cups, $N \geq 2$. For $k \geq 2$, the event $N = k$ happens exactly when Jim guesses correctly with $2, 3, \ldots, k-1$ cups and then fails with k cups. By independence,

$$\begin{aligned} \mathrm{P}(N = k) &= \frac{1}{2} \cdot \frac{1}{3} \cdots \frac{1}{k-1} \cdot \left(1 - \frac{1}{k}\right) = \frac{1}{(k-1)!} \cdot \frac{k-1}{k} \\ &= \frac{k-1}{k!}, \qquad k = 2, 3, 4, \ldots \end{aligned}$$

(For $k = 2$ the product of correct guesses is empty and we get $\frac{1}{2}$, as we should.)

It is worth noticing a neat equivalent form: $N > k$ means Jim was right with $2, \ldots, k$

cups, so

$$\mathrm{P}(N > k) = \frac{1}{k!} \qquad (k \geq 1).$$

Indeed $\frac{1}{(k-1)!} - \frac{1}{k!} = \frac{k-1}{k!}$, matching the formula above. As $k \to \infty$ this tends to 0, so the game ends with probability 1 and the probabilities sum to 1.

The expectation

Cancelling k against $k!$,

$$\begin{aligned}\mathrm{E}(N) &= \sum_{k=2}^{\infty} k \cdot \frac{k-1}{k!} = \sum_{k=2}^{\infty} \frac{k(k-1)}{k!} = \sum_{k=2}^{\infty} \frac{1}{(k-2)!} \\ &= \sum_{j=0}^{\infty} \frac{1}{j!} = \mathrm{e},\end{aligned}$$

putting $j = k-2$ and using $\mathrm{e}^x = \sum_{j \geq 0} x^j / j!$ at $x = 1$. This is the required result.

The variance

> **Ask yourself** Which second moment is easiest to sum here: $\mathrm{E}(N^2)$ or $\mathrm{E}\big(N(N-1)\big)$?

Either works, because the factor $k(k-1)$ in $k^2\,\mathrm{P}(N=k) = k \cdot \dfrac{k(k-1)}{k!}$ already cancels cleanly. We find

$$\begin{aligned}\mathrm{E}(N^2) &= \sum_{k=2}^{\infty} \frac{k^2(k-1)}{k!} = \sum_{k=2}^{\infty} \frac{k}{(k-2)!} = \sum_{j=0}^{\infty} \frac{j+2}{j!} \\ &= \sum_{j=1}^{\infty} \frac{1}{(j-1)!} + 2\sum_{j=0}^{\infty} \frac{1}{j!} = \mathrm{e} + 2\mathrm{e} = 3\mathrm{e},\end{aligned}$$

where in the middle step the $j = 0$ term of $\sum j/j!$ is zero, and $j/j! = 1/(j-1)!$ for $j \geq 1$. Therefore

$$\mathrm{Var}(N) = \mathrm{E}(N^2) - \big(\mathrm{E}(N)\big)^2 = 3\mathrm{e} - \mathrm{e}^2 = \mathrm{e}(3-\mathrm{e}) \approx 0.766.$$

Checks

- The probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{8}, \frac{1}{30}, \ldots$ add up to 1, as the tail form $\mathrm{P}(N > k) = 1/k!$ shows.
- Using the tail formula, $\mathrm{E}(N) = \sum_{k \geq 0} \mathrm{P}(N > k) = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots = \mathrm{e}$, an independent route to the same answer.
- The variance is positive, as it must be, since $\mathrm{e} < 3$. Exact sums give $\mathrm{E}(N) = 2.71828\ldots$ and $\mathrm{Var}(N) = 0.76579\ldots$, and a simulation of 400 000 games gave a mean of 2.720 and a variance of 0.767.

Common slips

- Starting the distribution at $k = 1$: there are two cups from the start, so $N \geq 2$.
- Forgetting the final factor $1 - \frac{1}{k}$ for the failed guess, which gives $1/(k-1)!$, and these "probabilities" sum to $\mathrm{e} - 1$, not 1.
- Writing $\mathrm{Var}(N) = \mathrm{E}(N^2) - \mathrm{E}(N)$ instead of subtracting $\big(\mathrm{E}(N)\big)^2$.

Beyond STEP

The tail-sum formula $\mathrm{E}(N) = \sum_{k \geq 0} \mathrm{P}(N > k)$, valid for any random variable taking values in $\{0, 1, 2, \dots\}$, is a standard first-year tool, and it often turns a messy expectation into a one-line calculation, as it did here. A cleaner way to package all the moments is the probability generating function $G(s) = \mathrm{E}(s^N)$. Here it works out to $G(s) = 1 + (s-1)\mathrm{e}^s$, from which $G'(1) = \mathrm{e}$ and $G''(1) = 2\mathrm{e}$ give the mean and variance at once. Grimmett and Welsh, *Probability: An Introduction*, develops generating functions in exactly this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.